

Digital Logic Design

Quiz 02

2025/03/17

姓名:

學號:

系級:

1. Simplify the Boolean expressions to a minimum number of literals: (18%)

- (a) $xy + xy'$
- (b) $(x + y)(x + y')$
- (c) $xyz + x'y + xyz'$
- (d) $(x + y)'(x' + y)'$
- (e) $(x + y + z')(x'y' + z)$
- (f) $x'yz + xyz' + xyz + x'yz'$

(a) $xy + xy' = x(y + y') = x$

(b) $(x + y)(x + y') = x + yy' = x(x + y') + y(x + y') = xx + xy' + xy + yy' = x$

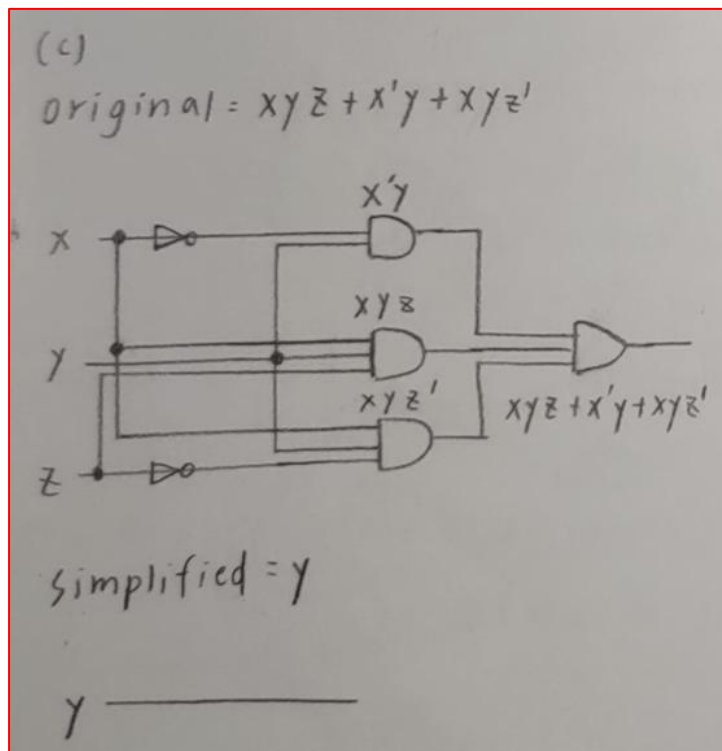
(c) $xyz + x'y + xyz' = xy(z + z') + x'y = xy + x'y = y$

(d) $(x + y)'(x' + y')' = (x'y')(xy) = (x'y')(yx) = x'(y'y)x = 0$

(e) $(x + y + z')(x'y' + z) = xx'y' + xz + ya'y' + yz + z'x'y' + z'z = xz + yz + x'y'z'$

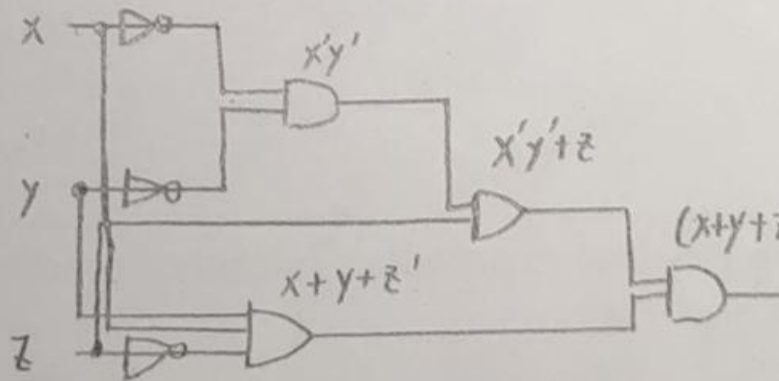
(f) $x'yz + xyz' + xyz + x'yz' = x'y(z + z') + xy(z + z') = x'y + xy = (x' + x)y = y$

2. Draw logic diagrams of the circuits that implement the original and simplified expression in Problem 1 (c), (e), and (f). (12%)

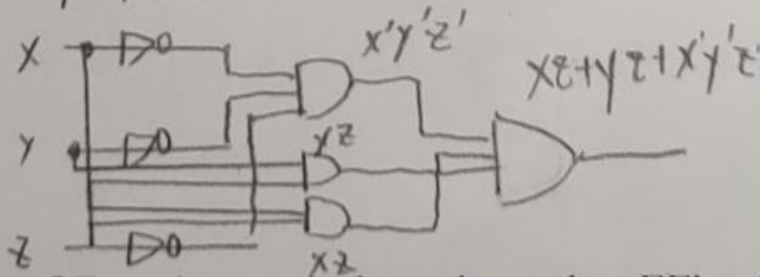


(e)

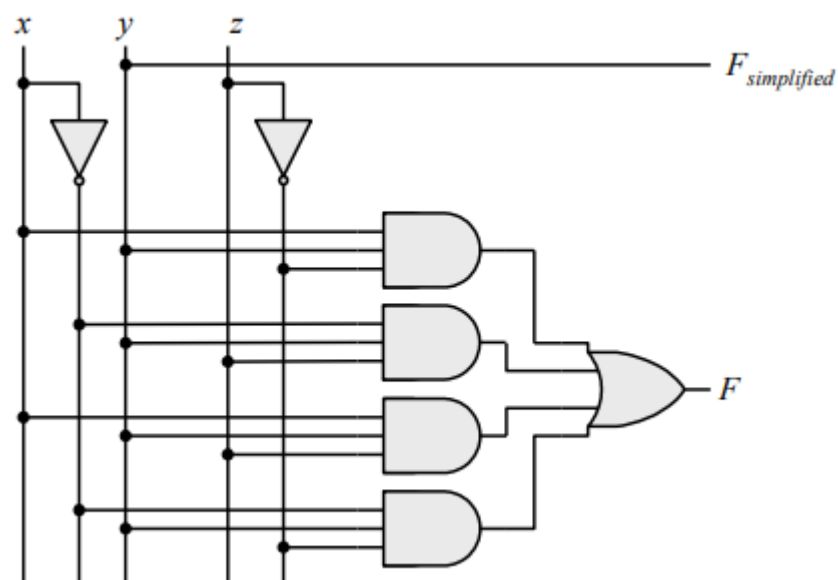
$$\text{original} = (x+y+z')(x'y'+z)$$



$$\text{Simplified} = xz + yz + x'y'z'$$



(f)



3. Find the complement of $F = x'y + yz'$; then show that $FF' = 0$ and $F + F' = 1$.(10%)

$$\begin{aligned}
 2.8 \quad F &= x'y + yz' \\
 F' &= (x'y + yz')' = (x'y)'(yz')' = (x + y')(y' + z) \\
 &= xy' + xz + y' + y'z = y'(x + z + 1) + xz = xz + y' \\
 FF' &= (x'y + yz')(y' + xz) = 0 \\
 F + F' &= x'y + yz' + y' + xz = y' + x'y + yz' + xz = y' + x' + xz + yz' \\
 &= y' + x' + z + yz' = y' + x' + z + y = 1
 \end{aligned}$$

4. Find the complement of the following expressions:(15%)

- (a) $xyz + x'y'z'$
 (b) $(x + y')(y + z')(z + x')$
 (c) $w'x' + w(x + y + z)$

$$\begin{aligned}
 2.9 \quad (a) \quad F' &= (xyz + x'y'z')' = (xyz)'(x'y'z')' = (x' + y' + z')(x + y + z) \\
 (b) \quad F' &= [(x + y')(y + z')(z + x')] = (x + y')' + (y + z')' + (z + x')' = x'y + y'z + z'x \\
 (c) \quad F' &= [w'x' + w(x + y + z)]' = (w'x')'[w(x + y + z)]' = (w + x)[w' + (x + y + z)] \\
 &= (w + x)(w' + x'y'z')
 \end{aligned}$$

5. We can perform logical operations on strings of bits by considering each pair of corresponding bits separately (called bitwise operation). Given two eight-bit strings $A = 11001010$ and $B = 10010011$, evaluate the eight-bit result after the following logical operations:(15%)

- (a) AND (b) OR (c) XOR (d) NOT A (e) NOT B

$$\begin{aligned}
 2.12 \quad A &= 1100_1010 \\
 B &= 1001_0011 \\
 (a) \quad A \text{ AND } B &= 1000_0010 \\
 (b) \quad A \text{ OR } B &= 1101_1011 \\
 (c) \quad A \text{ XOR } B &= 0101_1001 \\
 (d) \quad \text{NOT } A &= 0011_0101 \\
 (e) \quad \text{NOT } B &= 0110_1100
 \end{aligned}$$

6. Determine whether the following Boolean equation is true or false.(10%)

$$a'd' + b'c' + c'd' = (a' + c')(b' + d')(c' + d')$$

$$\begin{aligned} 2.29 \quad \text{L.H.S.} &= a'd' + b'c' + c'd' \\ &= (a' + c')d' + b'c' = (b'c' + d')(b'c' + a' + c') \\ &= (b' + d')(c' + d')(a' + c'(b' + 1)) \\ &= (a' + c')(b' + d')(c' + d') = \text{R.H.S.} \end{aligned}$$

Hence the equation is true.

7. Write the following Boolean expressions in sum of products form:(10%)

$$(w + x + y)(x' + y' + z')$$

- 2.30 Write the following Boolean expressions in sum of products form:

$$\begin{aligned} F &= (w + x + y)(x' + y' + z') \\ &= wx' + wy' + wz' + xx' + xy' + xz' + x'y + yy' + y'z' \\ &= wx' + wy' + wz' + xy' + xz' + x'y + y'z' \end{aligned}$$

8. Write the following Boolean expressions in products of sum form:(10%)

$$ab + bc + a'b'c'$$

$$\begin{aligned} 2.31 \quad F &= ab + bc + a'b'c' \\ &= (ab + b)(ab + c) + a'b'c' \\ &= (a + b)(b + b)(a + c)(b + c) + a'b'c' \\ &= (a + b + a'b'c')(b + a'b'c')(a + c + a'b'c')(b + c + a'b'c') \\ &= (a + b + a')(a + b + b')(a + b + c')(b + a')(b + b')(b + c')(a + c + a') \\ &\quad (a + c + b') \quad (a + c + c')(a' + b + c)(b + c + b')(b + c + c') \\ &= (a' + b)(b + c')(a + b + c')(a + b' + c)(a' + b + c) \end{aligned}$$